Lightweight Restraints for Powerful Agents: Effective AI Control Monitors at a Fraction of the Cost

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Abstract

[TODO: Abstract: problem (monitoring on the critical path of agent actions), method (binary-logit distillation from a frontier teacher into small open-weight monitors), headline results, and what is released.]

1 Introduction

[TODO: Motivate cheap, fast monitors: the safety tax of frontier monitors on the critical path of agent actions.]

[TODO: State contributions as a short list: training recipe, evaluation, analysis, released models and serving stack.]

2 Related work

[TODO: Trained control monitors (prior work before ours), rationale distillation, white-box probes, monitor-directed attacks [Terekhov et al., 2025].]

3 Training and serving Gleipnir monitors

[TODO: Task formulation: trajectory in, single decision token out.]

3.1 Binary-logit distillation

[TODO: Teacher probability on the decision token as a soft target; contrast with severity scores and rationale SFT.]

3.2 Data and training setup

[TODO: Data sources and grouped holdouts, LoRA configuration, model sizes, checkpoint averaging.]

3.3 Serving and cost

[TODO: Main serving decisions: single prefill pass, low-precision weights with QAD for the 4B, last-token final-layer attention, host-path changes. Cost per 1,000 evaluations at on-demand and committed-use B200 rates. Score change against BF16 serving. Released artifacts. Throughput per change goes in the appendix.]

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4 Main evaluation

[TODO: Metric definition (pAUROC@20 and supporting metrics), in-distribution and OOD suites, baselines.]

4.1 Cost–performance frontier

[TODO: Frontier figure and table with bootstrap intervals.]

5 Held-out and adversarial evaluation

[TODO: Held-out suites and SleightBench (stripped view); monitor-directed injections.]

6 Probing at half depth

[TODO: Hard-label then soft-target probes on a fixed probe family, compared with fine-tuning on quality and cost. Frame as representation analysis, not a deployment alternative.]

7 Limitations

[TODO: Scope of evaluation, teacher dependence, known failure modes and adversarial limits.]

8 Conclusion

[TODO: Summary and future work.]

Acknowledgments

Jeremias Ferrao received funding from BlueDot Impact's Rapid Grants Program. The grant paid for the API credits used to cache the Kimi K3 decision probabilities that form our training data. We also thank Neel Nanda for his support during the exploration phase of his MATS stream, which provided a stipend and a compute budget.

References

Mikhail Terekhov, Alexander Panfilov, Daniil Dzenhaliou, Caglar Gulcehre, Maksym Andriushchenko, Ameya Prabhu, and Jonas Geiping. Adaptive attacks on trusted monitors subvert AI control protocols. *arXiv preprint arXiv:2510.09462*, 2025.

A Additional results

[TODO: Per-source results, hyperparameters, prompts and reproducibility details.]